

BCSE303L

OPERATING SYSTEMS

Dr.N.Narayanan Prasanth

Associate Professor

Department of Database Systems

School of Computer Science and Engineering (SCOPE)

Vellore Institute of Technology, Vellore

Mail: **n.prasanth@vit.ac.in**

Cabin: SJT116 A40

Objectives

1. To introduce the operating system concepts, designs and provide skills required to implement the services.
2. To describe the trade-offs between conflicting objectives in large scale system design.
3. To develop the knowledge for application of the various design issues and services.

Course Outcomes

On completion of this course, student should be able to:

1. Interpret the evolution of OS functionality, structures, layers and apply various types of system calls of various process states.
2. Design scheduling algorithms to compute and compare various scheduling criteria.
3. Apply and analyze communication between inter process and synchronization techniques.
4. Implement page replacement algorithms, memory management problems and segmentation.
5. Differentiate the file systems for applying different allocation, access technique, representing virtualization and providing protection and security to OS.

SYLLABUS

Module:1	Introduction	3 hours
Introduction to OS: Functionality of OS - OS design issues - Structuring methods (monolithic, layered, modular, micro-kernel models) - Abstractions, processes, resources - Influence of security, networking, and multimedia.		
Module:2	OS Principles	4 hours
System calls, System/Application Call Interface – Protection: User/Kernel modes - Interrupts -Processes - Structures (Process Control Block, Ready List etc.), Process creation, management in Unix – Threads: User level, kernel level threads and thread models.		
Module:3	Scheduling	9 hours
Processes Scheduling - CPU Scheduling: Pre-emptive, non-pre-emptive - Multiprocessor scheduling – Deadlocks - Resource allocation and management - Deadlock handling mechanisms: prevention, avoidance, detection, recovery.		
Module:4	Concurrency	8 hours
Inter-process communication, Synchronization - Implementing synchronization primitives (Peterson's solution, Bakery algorithm, synchronization hardware) - Semaphores – Classical synchronization problems, Monitors: Solution to Dining Philosophers problem – IPC in Unix, Multiprocessors and Locking - Scalable Locks - Lock-free coordination.		

SYLLABUS

Module:5	Memory Management	7 hours
Main memory management, Memory allocation strategies, Virtual memory: Hardware support for virtual memory (caching, TLB) – Paging - Segmentation - Demand Paging - Page Faults - Page Replacement -Thrashing - Working Set.		
Module:6	Virtualization and File System Management	6 hours
Virtual Machines - Virtualization (Hardware/Software, Server, Service, Network - Hypervisors - Container virtualization - Cost of virtualization - File system interface (access methods, directory structures) - File system implementation (directory implementation, file allocation methods) - File system recovery - Journaling - Soft updates - Log-structured file system - Distributed file system.		
Module:7	Storage Management, Protection and Security	6 hours
Disk structure and attachment – Disk scheduling algorithms (seek time, rotational latency based)- System threats and security – Policy vs mechanism - Access vs authentication -		

Module 8: Contemporary Issues

2 hours

Text Books

1. Abraham Silberschatz, Peter B. Galvin, Greg Gagne, "Operating System Concepts", 2018, 10th Edition, Wiley, United States.

Reference Books

1. Andrew S. Tanenbaum, "Modern Operating Systems", 2016, 4th Edition, Pearson, United Kingdom.

2. William Stallings, "Operating Systems: Internals and Design Principles", 2018, 9th Edition, Pearson, United Kingdom.

Course Assessment Details

1. Digital Assignment 1 – 10 Marks (Before FAT)
2. Technical Quiz 1 – 10 Marks (Before CAT 1)
3. Technical Quiz 2 – 10 Marks (Before CAT 2)
4. CAT 1 – 15 Marks (will be scheduled by university)
5. CAT 2 – 15 Marks (will be scheduled by university)
6. FAT – 40 Marks (will be scheduled by university)

Module – 1

Introduction to Operating Systems

- ❑ Operating System is a **System Software**.
- ❑ It is designed to manage the computer hardware and provide a *platform* for running application software.
- ❑ It is a software that acts as an *interface/intermediary* between computer hardware components and the user.
- ❑ It provides the environment within which programs are executed.
- ❑ It performs tasks in a wide variety of *computing environments* such as smart phones, personal computers, enterprise computers, home appliances and many more.

Goals of OS

1. To execute user programs and solve user problems easier
2. To make the computer system convenient to use
3. To use the computer hardware in an efficient manner
4. To provide a user-friendly and convenient environment for the user
5. To fully utilize the computing resources

What OS do?

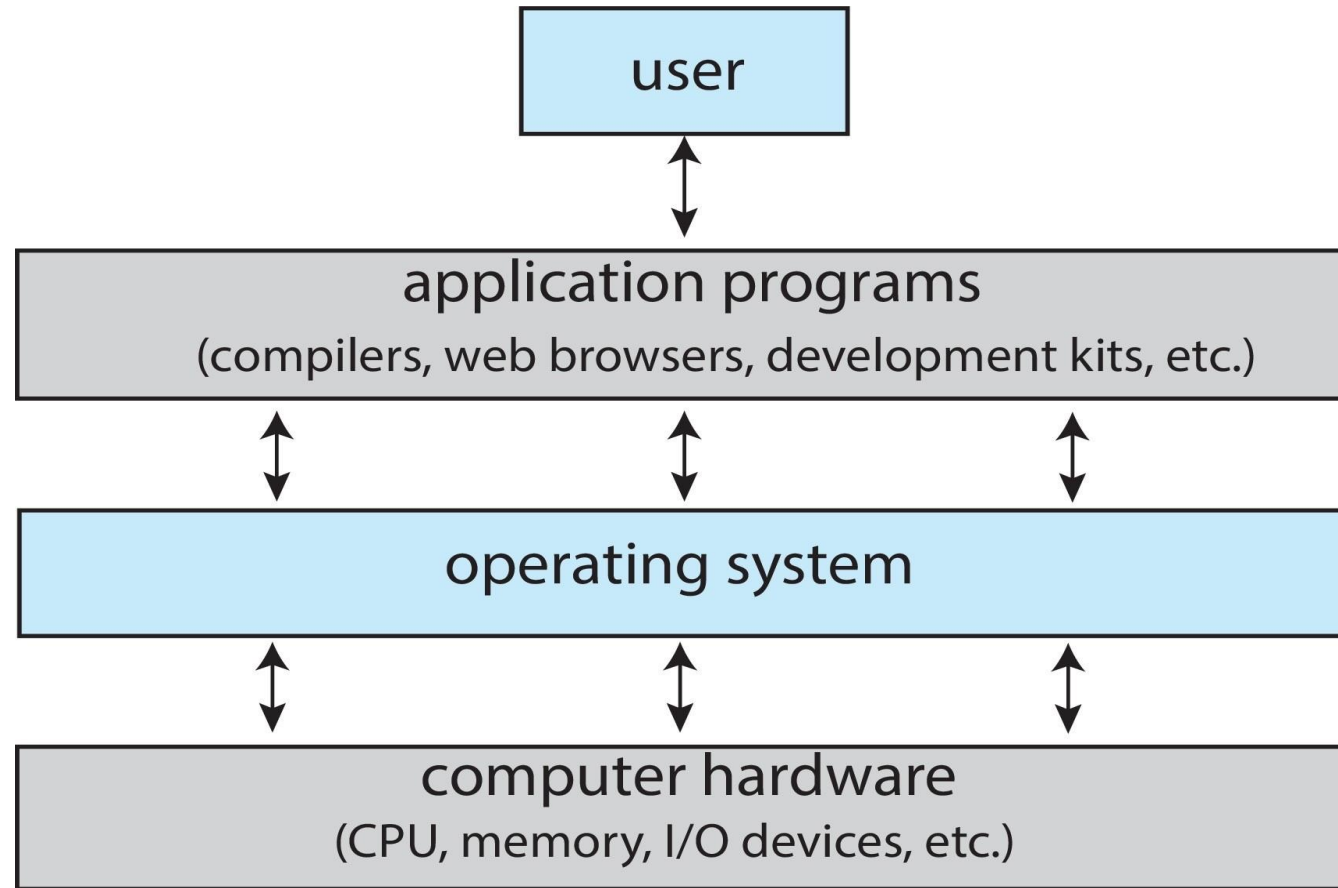


Figure 1 Abstract view of the components of a computer system

OS usage

- Hardware Abstraction
turns hardware into something that applications can use
- Resource Management
manage system's resources

What OS do?

It **controls the hardware** and coordinates its use among the various application programs for the various users.

*Explore operating systems from two viewpoints: **user and system***

User View: The user's view of the computer varies according to the interface being used.

Example: Single user i.e. laptop user and multiple user

(importance given to ease of use, performance and security but not to resource utilization).

What OS do?

System View:

It act as an Resource Allocator

It act as a *manager for many resources* such as CPU time, memory space, storage space, I/O devices, and many more.

It decide *how to allocate* them to specific programs and users so that it can operate the computer system efficiently and fairly.

It emphasizes the need to control the various I/O devices and user programs.

An operating system is a control program

A control program manages the execution of user programs to prevent errors and improper use of the computer. It is especially concerned with the operation and control of I/O devices.

An Example

A Simple Program

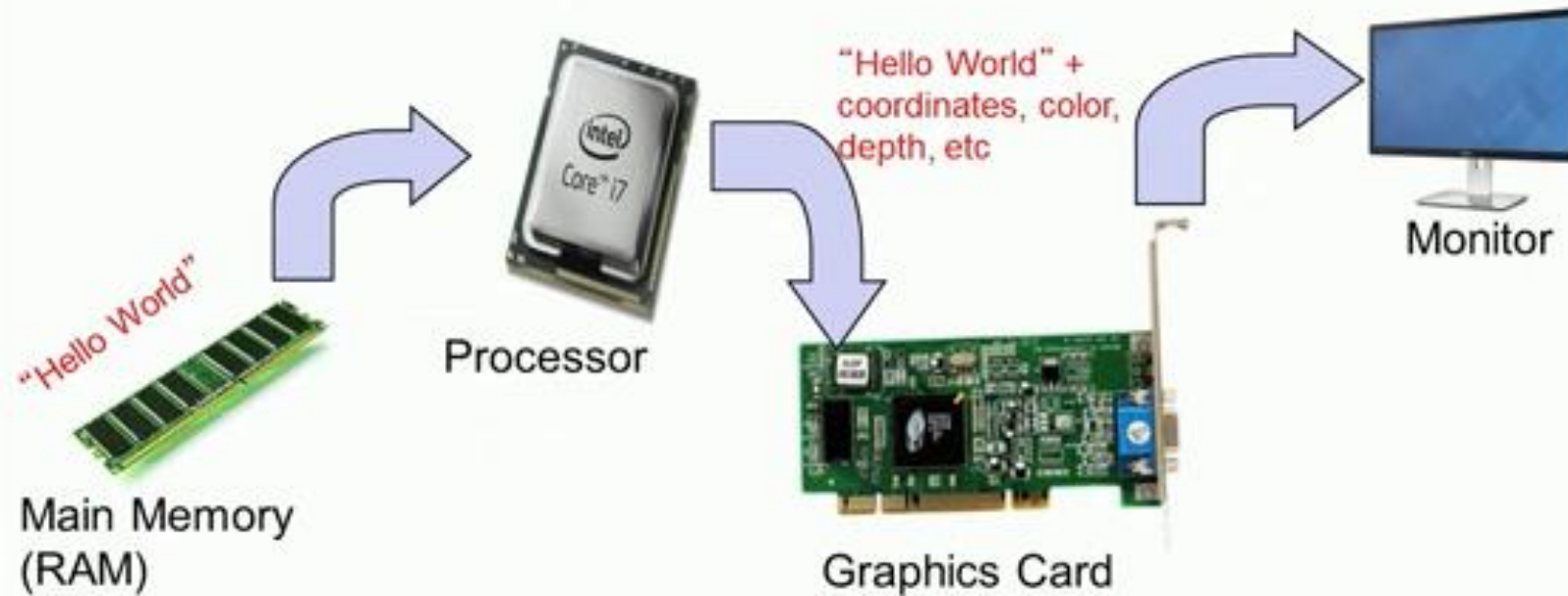
What is the output of the following program?

```
#include <stdio.h>

int main(){
    char str[] = "Hello World\n";
    printf("%s", str);
}
```

How is the string displayed on the screen?

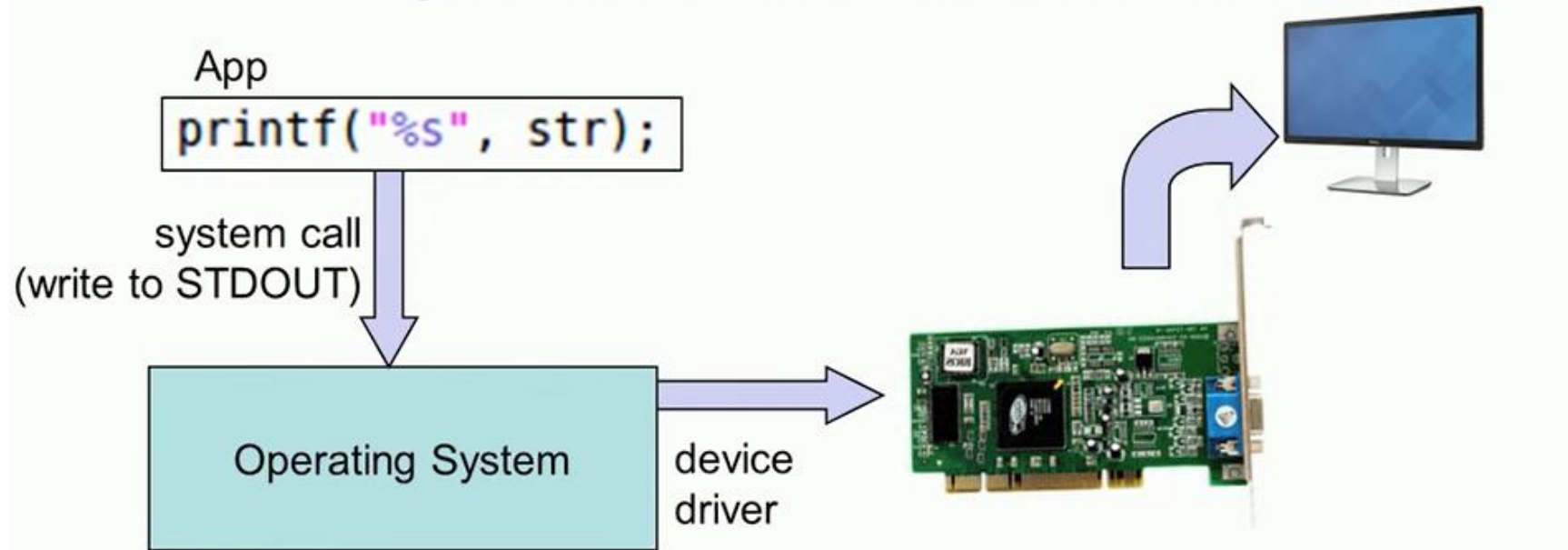
Displaying on the Screen



- Can be complex and tedious
- Hardware dependent

Without an OS, all programs need to take care of every nitty gritty detail

Operating Systems provide **Abstraction**



- **Easy to program apps**
 - No more nitty gritty details for programmers
- **Reusable functionality**
 - Apps can reuse the OS functionality
- **Portable**
 - OS interfaces are consistent. The app does not change when hardware changes

OS as a **Resource Manager**

- Multiple apps but limited hardware

Apps



```
int main() {  
    char str[] = "Hello World";  
    printf("%s", str);  
}
```



Operating Systems



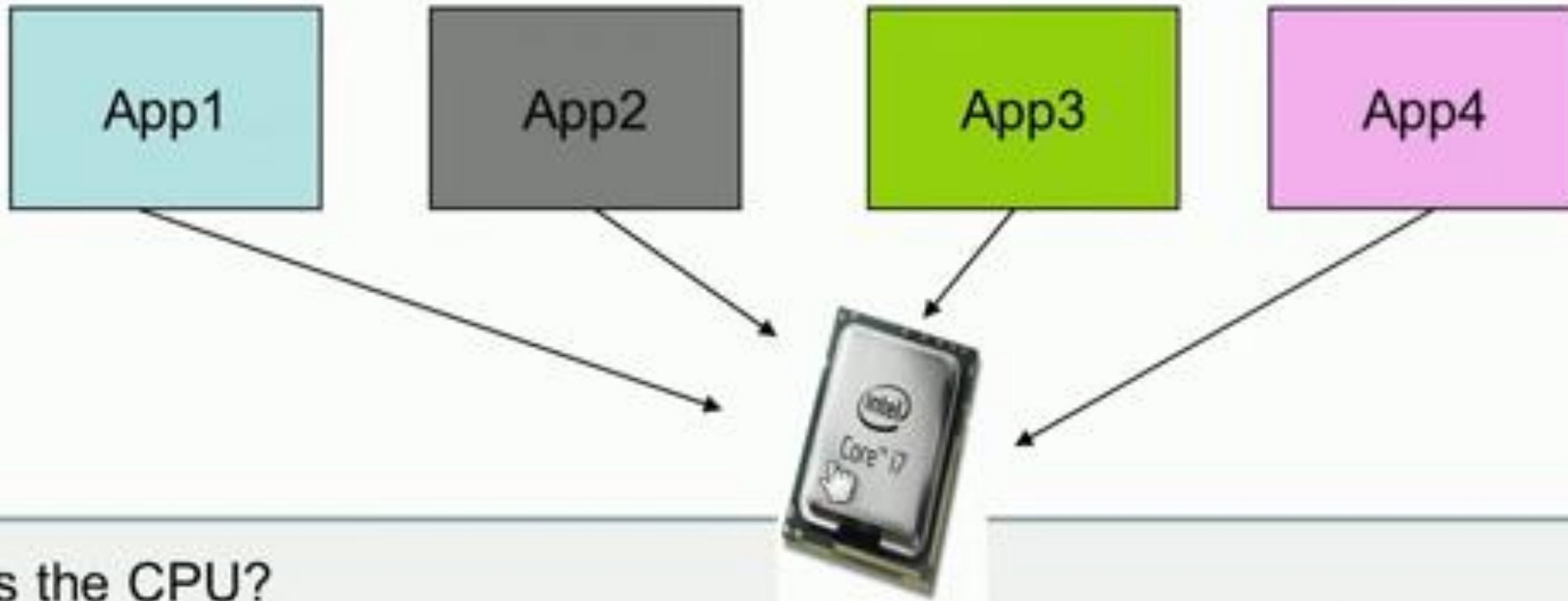
Allows sharing of hardware!!

OS as Resource Manager

- OS must manage CPU, memory, network, secondary storage (hard disk), etc...
- Resource management
 - allows multiple apps to share resources
 - protects apps from each other
 - Improves performance by efficient utilization of resources

For example,

Sharing the CPU



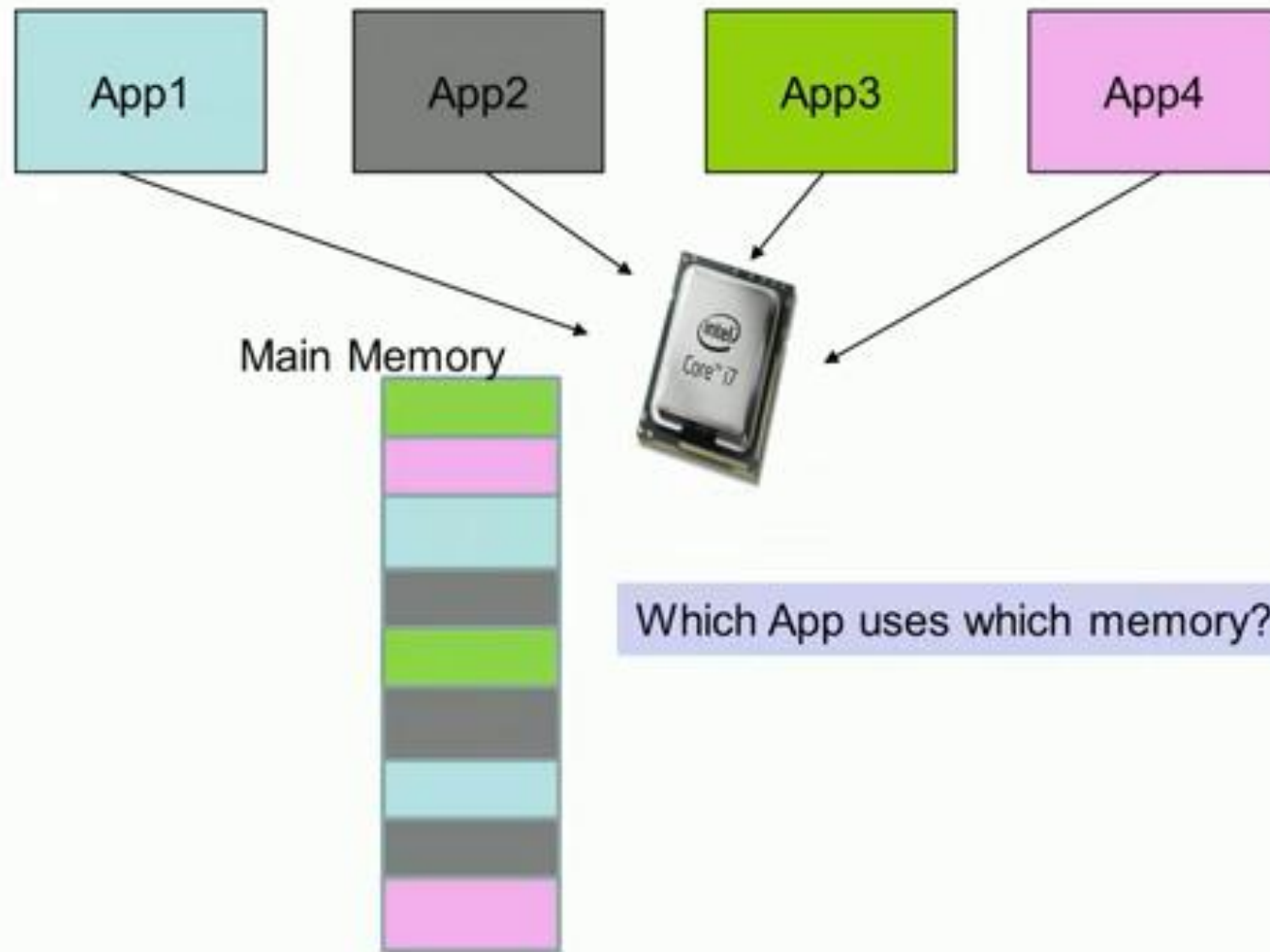
Who uses the CPU?



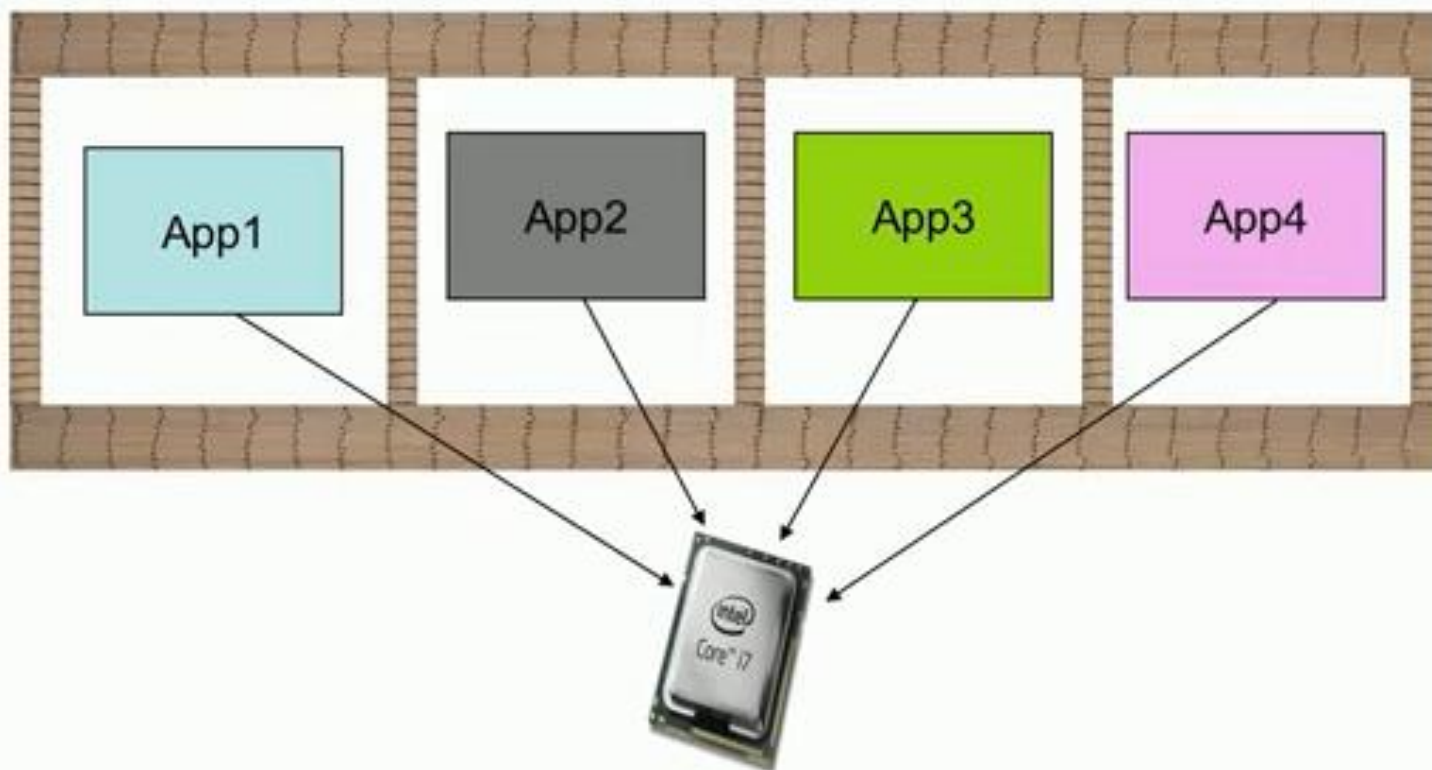
time



Sharing Memory



Share but Isolate



Share resources but keep applications isolated from each other

What OS do?

- *Perform Resource Allocation.*
- *Control all the Programs running in their environment.*
- *Facilitate resource sharing.*
- *Facilitate optimization of programs.*
- *Facilitates different user interfaces.*
- *Makes computer systems to be used easily and with needed performance.*

Resources managed by OS

- Central Processing Unit
- Main memory
- Secondary storage devices
- Input-Output devices
- Network connections
- Peripheral devices

Operating-System Services / Functionalities

- An operating system provides an environment for the execution of programs.
- The specific services provided can differ from one operating system to another.

*One set of operating system services provides functions that are **helpful to the user**.*

1. User Interface: Almost all operating systems have a user interface (UI). They are in different forms.

For example: command-line interface, batch interface, GUI

Operating-System Services / Functionalities

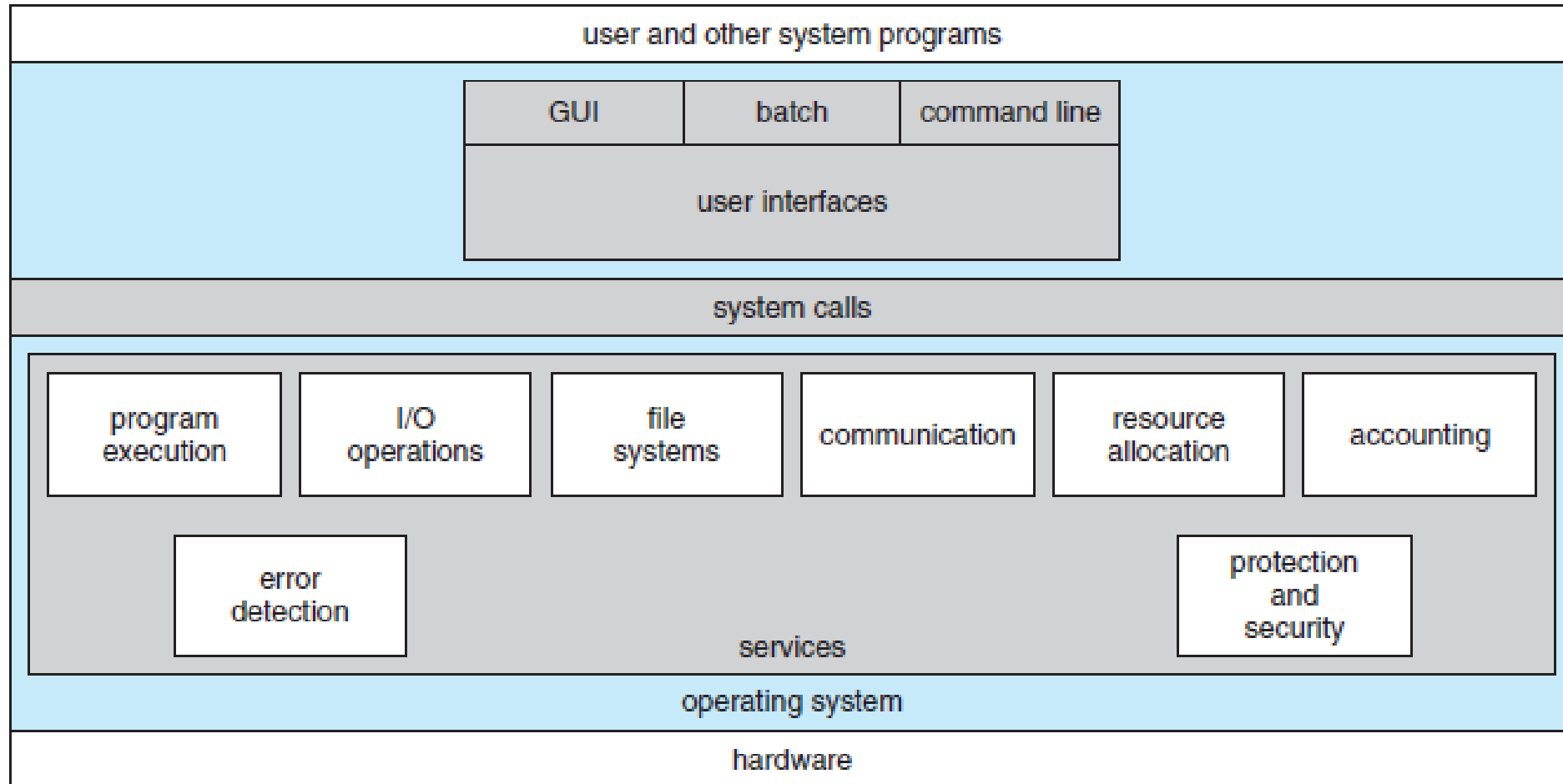


Figure: A view of operating system services

Operating-System Services

2. Program execution - The system must be able to load a program into memory and to run that program, end execution, either normally or abnormally.

3. I/O operations - A running program may require I/O, which may involve a file or an I/O device.

4. File-system manipulation - The file system is of particular interest. Programs need to read and write files and directories, create and delete them, search them, list file Information, permission management.

Operating-System Services

5. Communication – Process to process communication located on the same or different computer. *Example approaches: Shared memory, message passing*

6. Error detection – OS needs to be constantly aware of possible errors.

- *May occur in the CPU and memory hardware, in I/O devices, in user program*
- For each type of error, OS should take the appropriate action to ensure correct and consistent computing.
- Debugging facilities can greatly enhance the user's and programmer's abilities to efficiently use the system.

Operating-System Services

Another set of operating system functions exists *not for helping the user but rather for ensuring the efficient operation of the system*. Systems with multiple users can gain efficiency by sharing the computer resources among the users.

1. Resource allocation: When there are multiple jobs running at the same time, resources must be allocated to each of them. Some resources are allocated based on scheduling schemes and some are allocated directly.

Operating-System Services

2. Accounting: Keep track of resource usage details of all the users. This record may be used for accounting (so that users can be billed) or simply for accumulating usage statistics.

3. Protection and security. The owners of information stored in a multiuser or networked computer system may want to control use of that information.

- Protection involves ensuring that all access to system resources is controlled.
- Security starts with *user authentication* to gain access to system resources.
- Protect devices from invalid access attempts.

Operating-System Services

Some of the other services are:

- 1. Process management:** The operating system is responsible for starting, stopping, and managing processes and programs. It also controls the scheduling of processes and allocates resources to them.
- 2. Memory management:** The operating system manages the computer's primary memory and provides mechanisms for optimizing memory usage.
- 3. Security:** The operating system provides a secure environment for the user, applications, and data by implementing security policies and mechanisms such as access controls and encryption.

Operating-System Services

- 4. **File management:** The operating system is responsible for organizing and managing the file system, including the creation, deletion, and manipulation of files and directories.
- 5. **Device management:** The operating system manages the communication between the computer system and its peripheral devices such as printers, displays, keyboards, and mice.
- 6. **Networking:** The operating system provides networking capabilities that allow the computer system to communicate with other systems and access remote resources.

References

1. **Abraham Silberschatz, Peter B. Galvin, Greg Gagne, “Operating System Concepts”, 2018, 10th Edition, Wiley, United States.**
2. **Introduction to Operating System – NPTEL Course.**